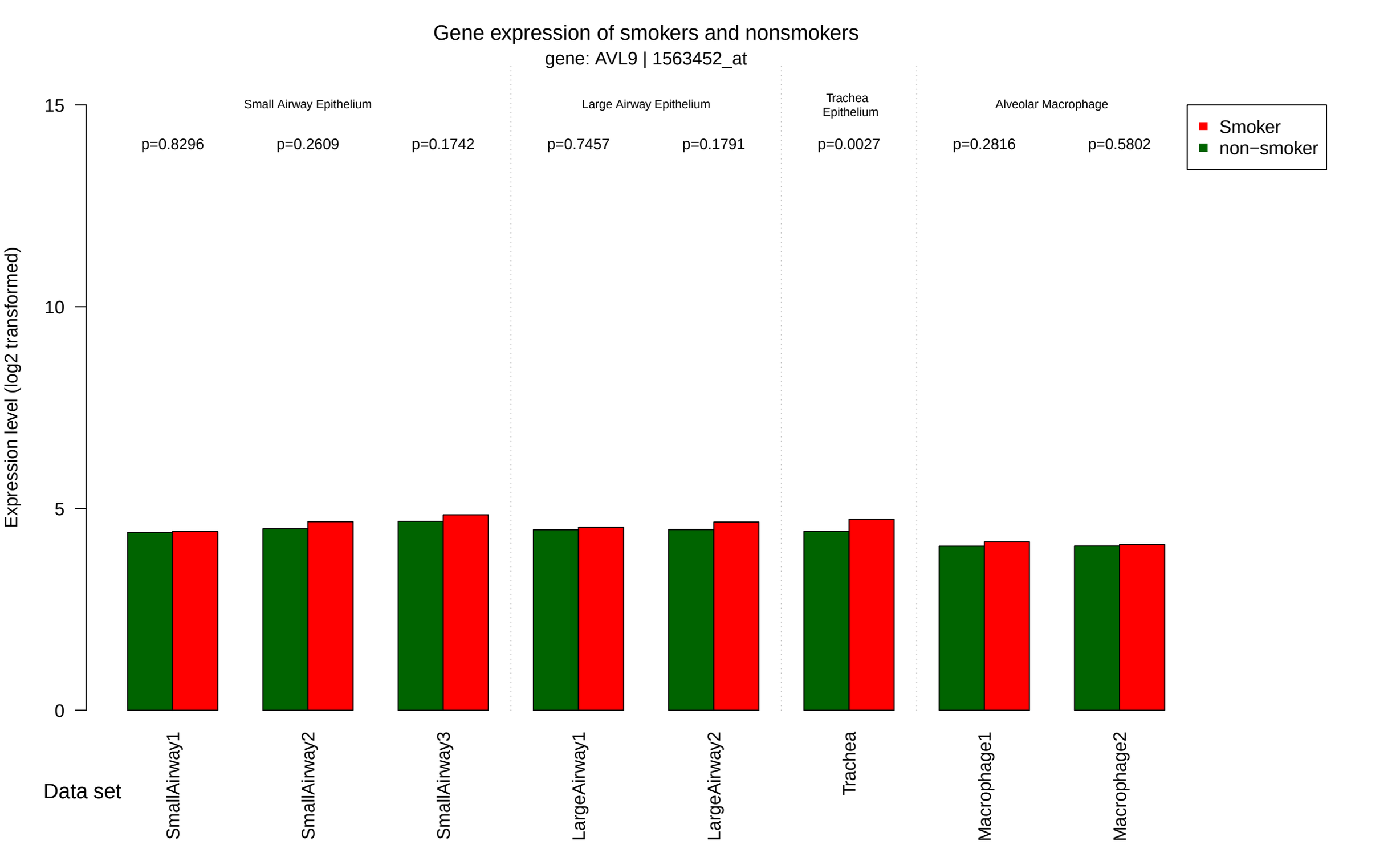
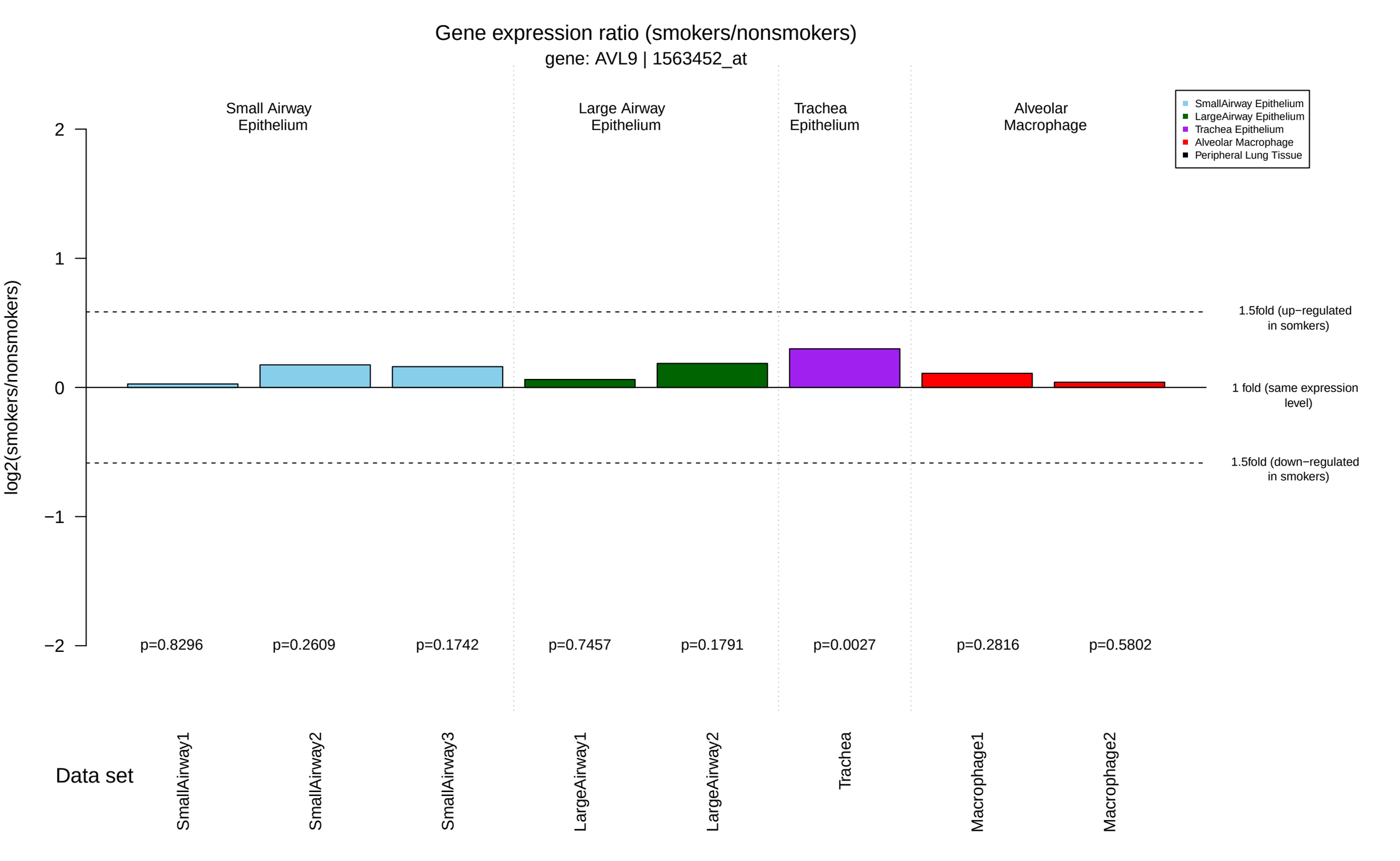
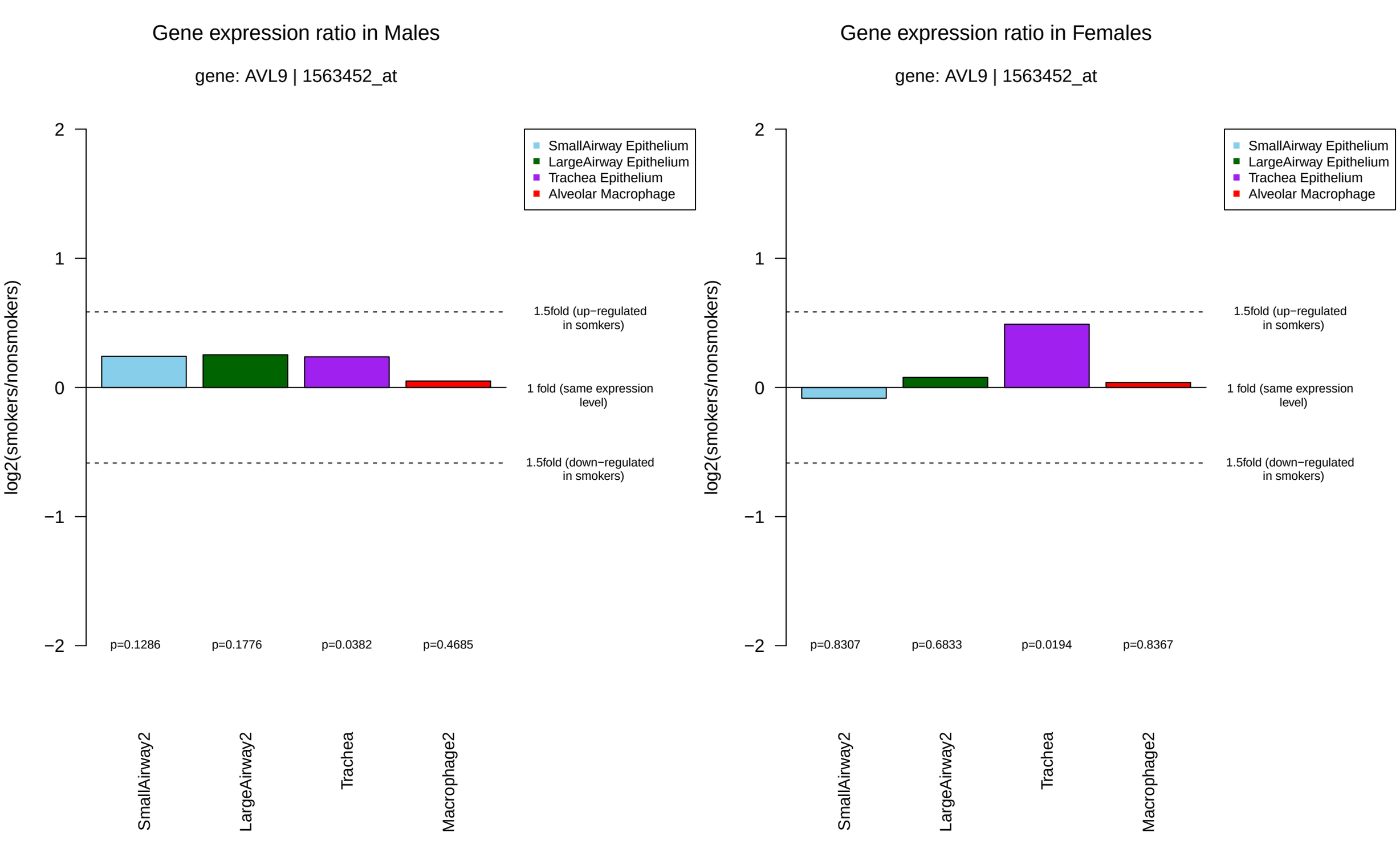
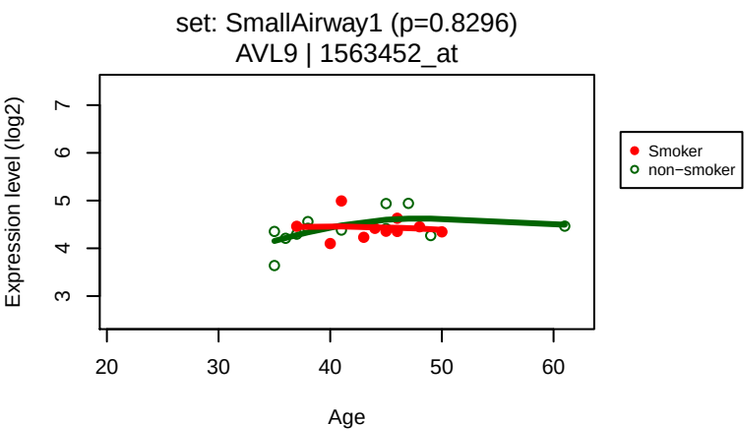
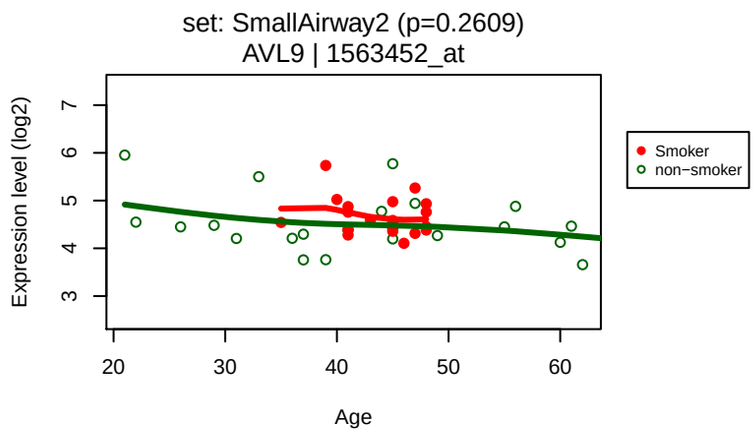
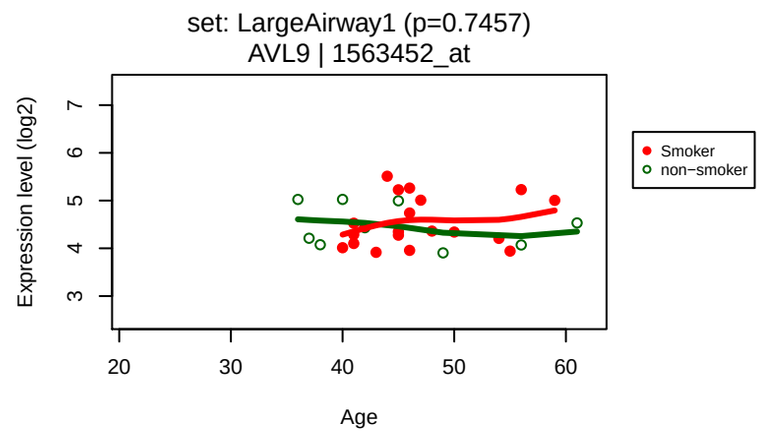

Gene expression of smokers and nonsmokers
gene: AVL9 | 1563452_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8296
p=0.2609
p=0.1742
p=0.7457
p=0.1791
p=0.0027
p=0.2816
p=0.5802
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

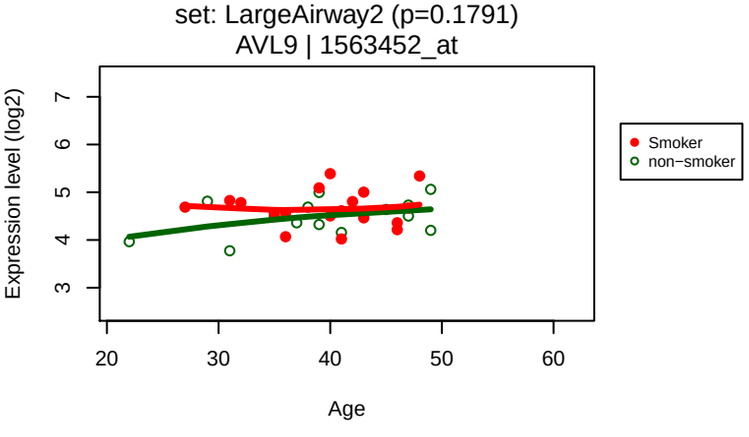

Gene expression ratio (smokers/nonsmokers)
gene: AVL9 | 1563452_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8296
p=0.2609
p=0.1742
p=0.7457
p=0.1791
p=0.0027
p=0.2816
p=0.5802
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

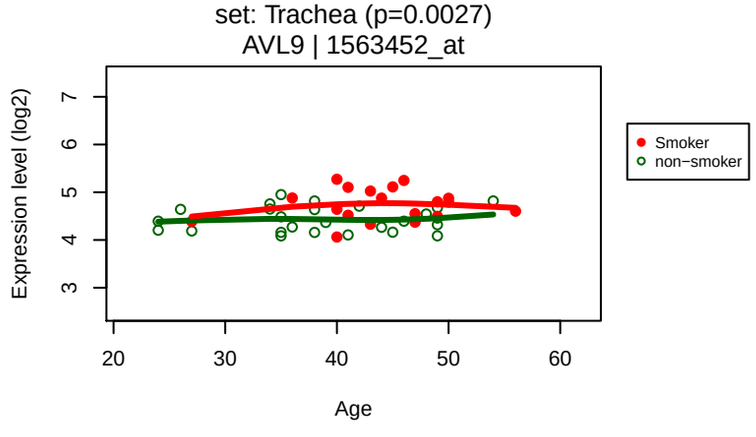

Gene expression ratio in Males
gene: AVL9 | 1563452_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1286
p=0.1776
p=0.0382
p=0.4685
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AVL9 | 1563452_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8307
p=0.6833
p=0.0194
p=0.8367
SmallAirway2
LargeAirway2
Trachea
Macrophage2

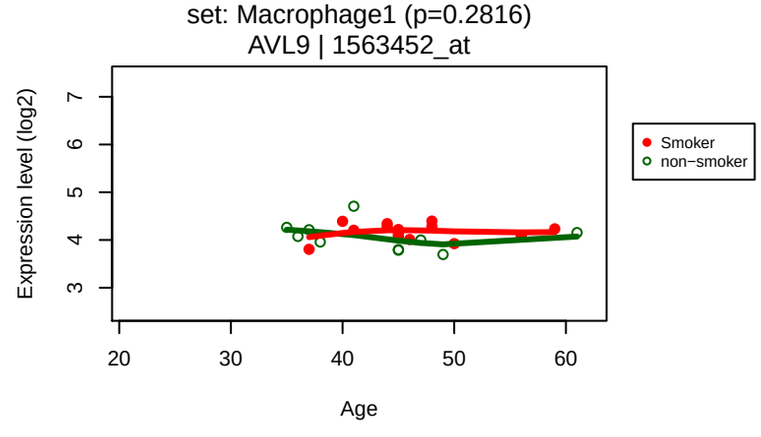

set: SmallAirway1 (p=0.8296)
AVL9 | 1563452_at
Expression level (log2)
Age
Smoker
non−smoker

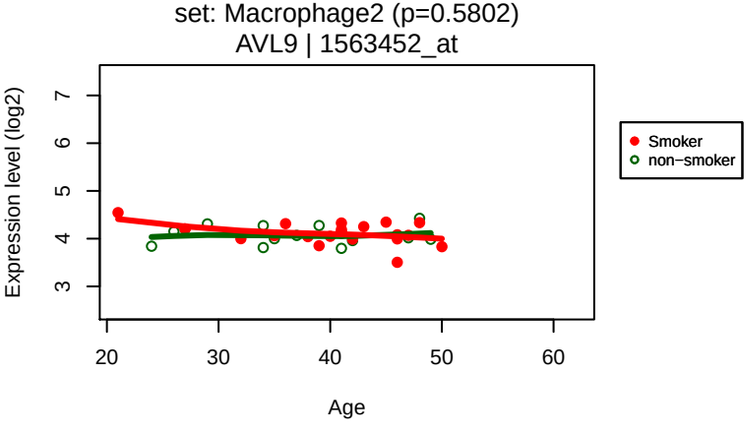

set: SmallAirway2 (p=0.2609)
AVL9 | 1563452_at
Expression level (log2)
Age
Smoker
non−smoker

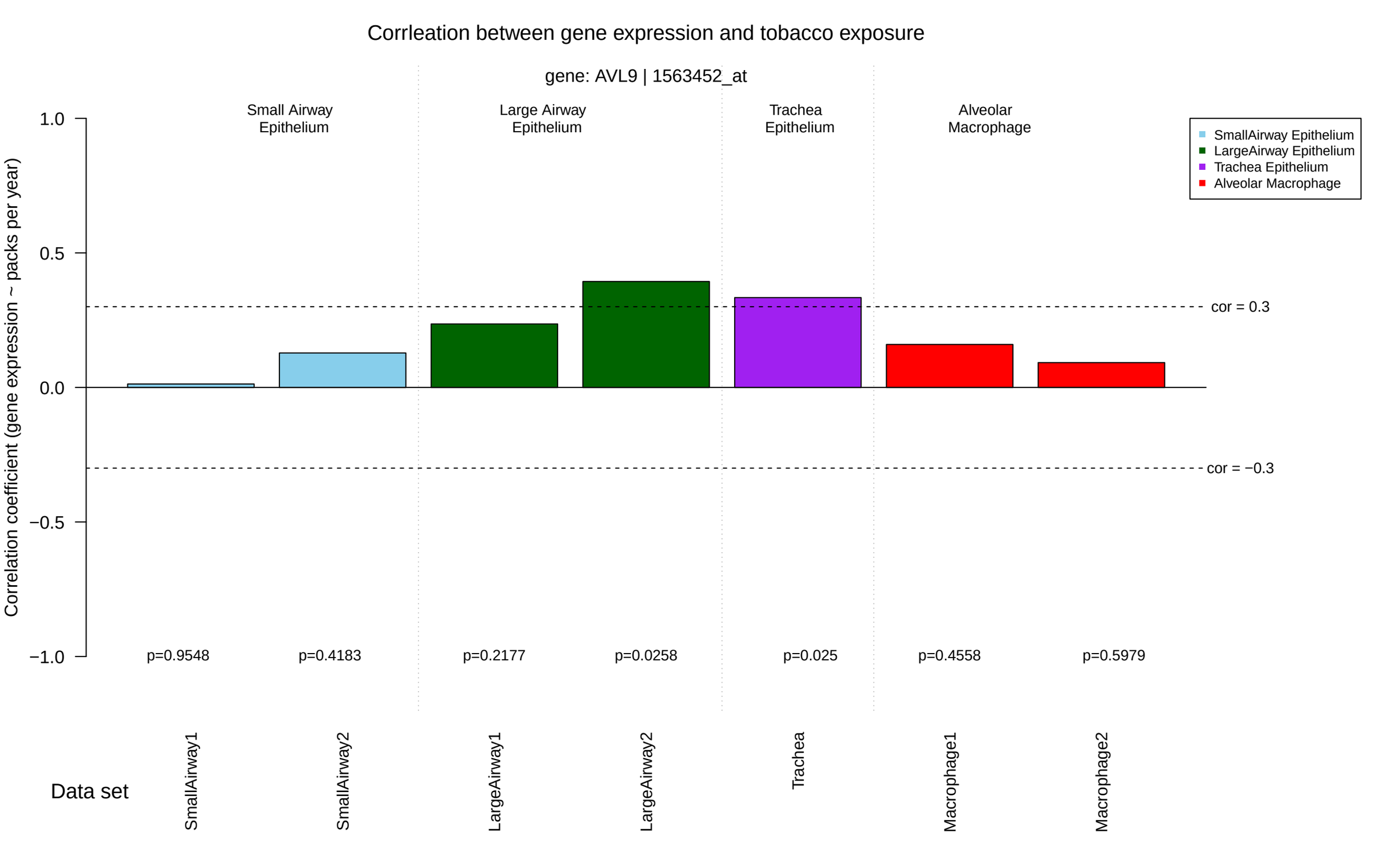

set: LargeAirway1 (p=0.7457)
AVL9 | 1563452_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1791)
AVL9 | 1563452_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0027)
AVL9 | 1563452_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2816)
AVL9 | 1563452_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5802)
AVL9 | 1563452_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AVL9 | 1563452_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9548
p=0.4183
p=0.2177
p=0.0258
p=0.025
p=0.4558
p=0.5979
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2